

EEE3094S 2022: Class Test 1

- The test is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages”) and make sure they are marked with your student number.
- There are four questions totalling 20 marks. There are also two bonus marks available, but the total may not exceed 100%.
- You have one hour to complete the test.
- The last page is an information sheet.
- The information sheet includes a photo of my housemate’s cat, Krummel. You are invited to draw a picture of him if you finish/ give up early and are bored. Doing so will not officially get you extra marks, but it will probably make tutor marking your test happy, which might make a difference in your favour.

Note: There are other ways to get to many of these solutions. Even if your method is different, you will still get the marks as long as what you did makes sense.

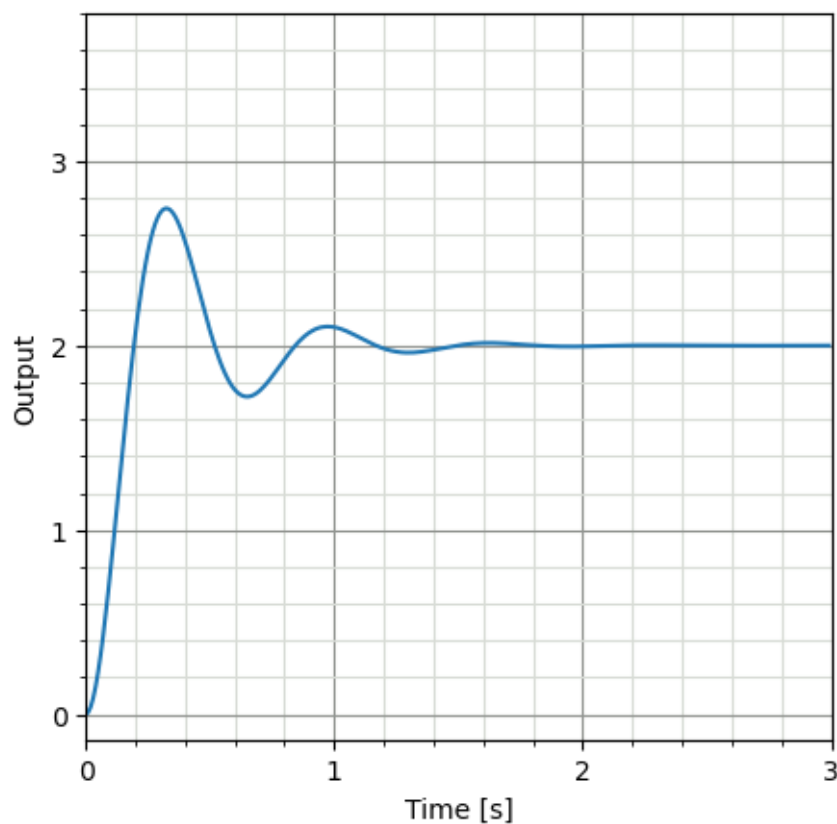
Question 1 [5]

Consider an unknown second-order system,

$$G(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The response of the system must satisfy the following specifications:

- Settling time to within 5% of the final value of less than 1 second.
 - Damping ratio greater than 0.5.
- a) Calculate the locations of the poles that satisfy these requirements with the minimum damping ratio. [2 marks]
- b) The unit step response of the system is shown below. Determine whether it satisfies these specifications. [3 marks]



Answer 1:

- a) Substituting a damping ratio of 0.5 into the settling time formula gives $\frac{\ln(0.05\sqrt{1-0.5^2})}{-0.5\omega_n} \leq 1$, so we can solve to find $\omega_n = 6.28$. [1]

From here, you can either substitute ω_n and ζ into the characteristic equation and find the roots, OR calculate the real ($-\zeta\omega_n$) and imaginary ($\omega_n\sqrt{1-\zeta^2}$) parts of the roots. Either approach gives poles at $3.14 \pm j5.44$. [1]

- b) Both specifications require the damping ratio to test, so we need to determine the damping ratio based on the step response. The damping ratio and gain are the only unknowns in the peak value formula.

The gain of $A = 2$ can be obtained from the final value of the response. [1]

The peak value is around 2.7. Substituting this and the gain into the formula gives

$$2.7 = 2 \left(1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}} \right), \text{ which can be solved for a damping ratio of } \zeta = 0.3. [1]$$

This does not satisfy the specifications. [1]

Question 2 [5]

A system with the open-loop transfer function $G(s) = \frac{1}{s+2}$ is placed in closed loop to improve its ability to track a step input.

Two possible designs are proposed to control the closed-loop system:

Option 1: a proportional controller $C_1(s) = K$

Option 2: an integral controller $C_2(s) = \frac{K}{s}$

The final design must be able to satisfy the following specifications:

- Greater than 95% error with respect to position (step) inputs.
 - Settling time not exceeding that of the open-loop system.
 - Damping ratio greater than 0.7
- a) Calculate the error constant of the uncontrolled closed-loop system with respect to a unit step input. [2 marks]
- b) Which of the controller options should be used? You may support your argument with calculations and/or rough sketches of the root locus. [3 marks]

Answer 2:

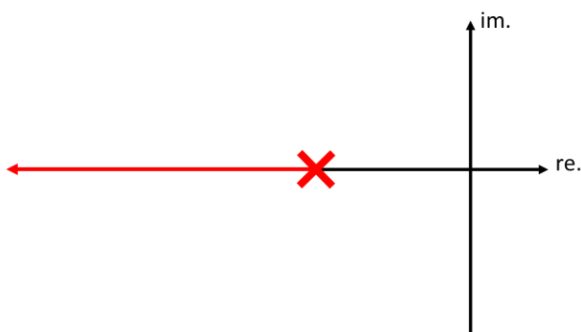
- a) The error is related to the setpoint by the transfer function $\frac{1}{1+G(s)}$. [1]

The error constant is given by the final value of this transfer function when a step input is applied:

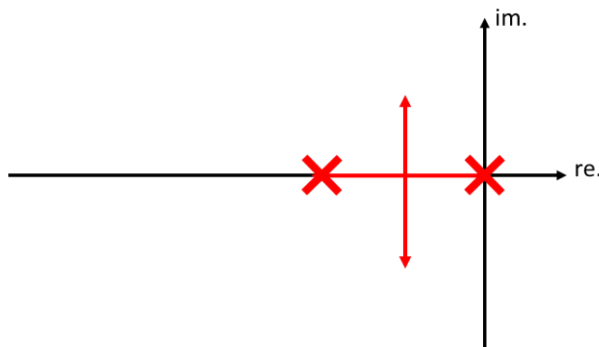
$$\lim_{s \rightarrow 0} s \frac{1}{1+G(s)} \frac{1}{s} = \frac{1}{1+\frac{1}{2}} = 0.67. [1]$$

- b) The open-loop gain needs to be increased to improve the steady-state tracking, so we must examine the root locus to determine whether the closed-loop poles will satisfy the requirements as the gain increases.

For option 1 (the proportional controller), there will still be one closed-loop pole, and increasing the gain will cause it to move leftwards away from the origin. The response at higher gain will therefore settle faster than the open-loop system. As the pole is real, its behaviour corresponds to a damping ratio > 1 . This option is therefore able to satisfy all requirements. [1]



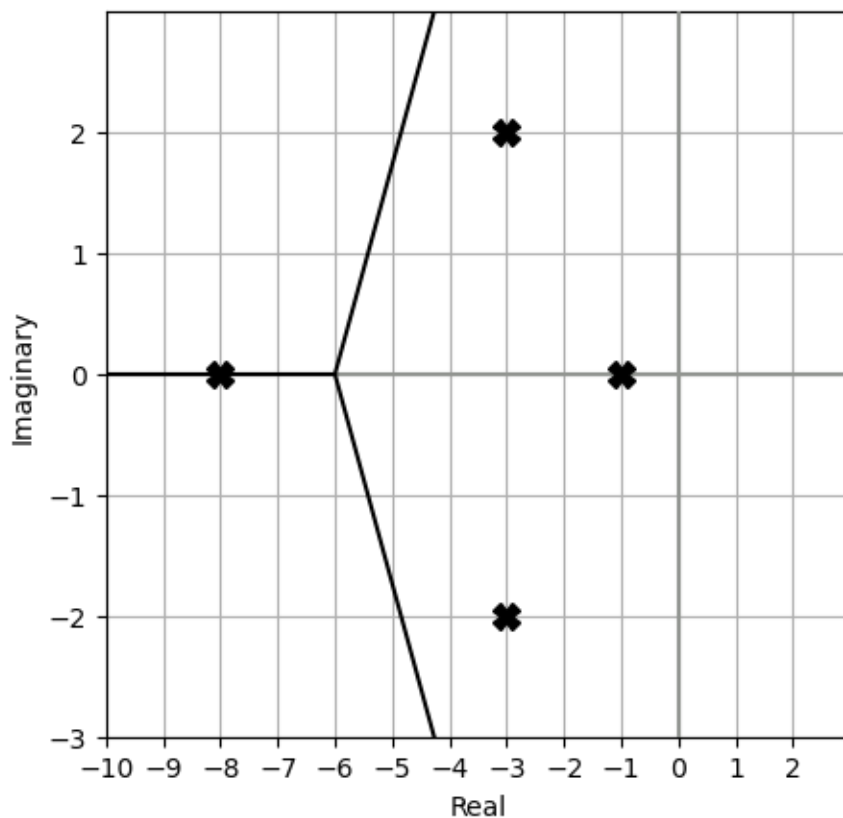
For option 2 (the integral controller), steady-state tracking will be perfect, but the additional pole at the origin means that all poles on the root locus will be slower than the open-loop pole. [1]



Option 1 is therefore the only viable option. [1]

Question 3 [5]

The plot below shows the open-loop poles of a system, and the asymptotes of its root locus (the black lines).



- Use this information to show that the system must have one zero located in the right-half plane at $s = 3$. [2 marks]
- True or false: the unit step response of the open-loop system will initially move away from the final value. Explain. [1 mark]
- Complete the root locus by showing where it exists on the real axis, and roughly sketching the trajectories of the poles. (You will need to redraw the plot.) [2 marks]

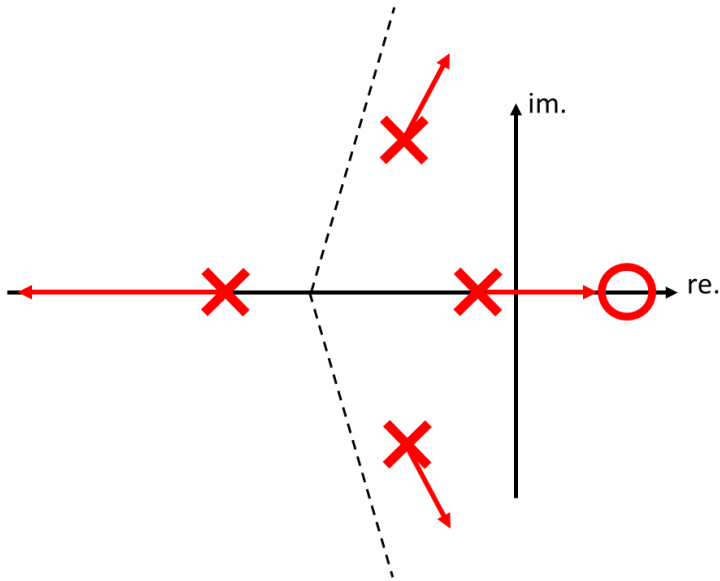
Answer 3:

- a) There are four poles and only three asymptotes. The number of asymptotes = n (poles) - m (zeros), so there must be at least one zero. [1]

The centroid of the asymptotes is located at -4. This is calculated using $\sigma_a = \frac{\sum_{i=1}^n p - \sum_{j=1}^m z}{m-n}$. The sum of the poles is -15, so the formula becomes $-4 = \frac{-15-z}{3}$, which can be solved to give a zero at 3. [1]

- b) True: the nonminimum phase (right-half plane) zero causes the response to move in the opposite direction to the final value.

c)



Question 4 [5]

A system with the open-loop transfer function $G(s) = \frac{1}{s+6}$ is required to track a velocity input (that is, a ramp) without error in closed loop.

Three controller designs are suggested:

Option 1: $C_1(s) = \frac{1}{s^2}$

Option 2: $C_2(s) = \frac{s+5}{s^2}$

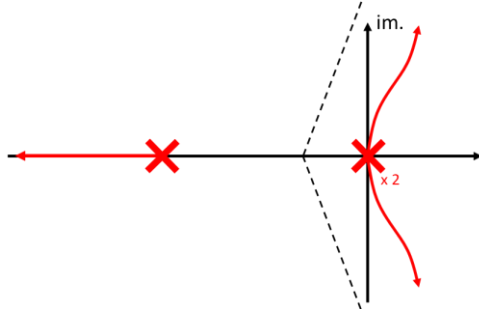
Option 3: $C_3(s) = \frac{s+7}{s^2}$

- a) With reference to the type number of the system, explain why the controller must add at least two poles at the origin. [1 mark]
- b) Only one of the proposed controllers will result in an acceptable closed-loop system. Which one is it? Support your answer with root locus sketches for the three designs. [4 marks]
- c) BONUS: True or false: an underdamped second-order model is an acceptable approximation of the complete system with the working design. Explain. [2 marks]

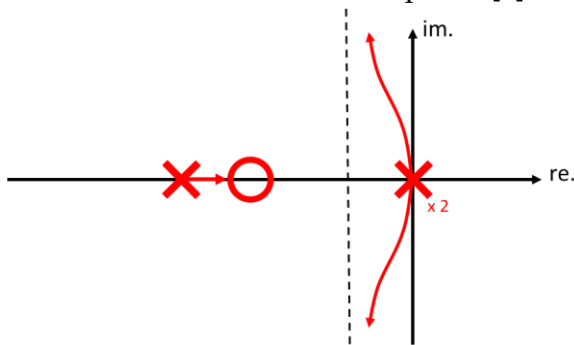
Answer 4:

- The type number gives the number of poles at the origin in the open-loop system. The type number must be greater than or equal to the order of the input, so a second-order input (a ramp) requires a type 2 system to track with perfect accuracy. [1]
- The poles for all three proposed designs will be the same, only the number and positions of the zeros changes. [1]

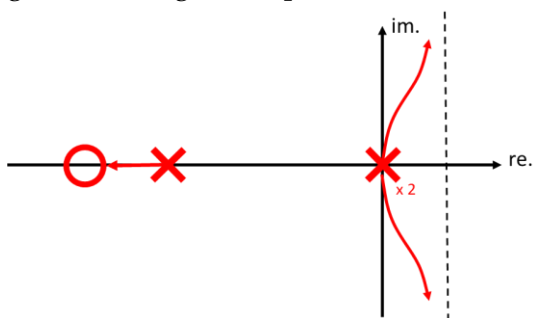
Option 1 is not acceptable, as the asymptotes show that the poles at the origin will become unstable in closed loop for any gain. [1]



Option 2 has the centroid of the asymptotes at $\frac{-6+0+0-(-5)}{2} = -0.5$, so the poles at the origin will be directed back into the left-half plane. [1]



Option 3 has the centroid of the asymptotes at $\frac{-6+0+0-(-7)}{2} = 0.5$, so the poles at the origin will still go into the right-half plane. [1]



Option 2 is therefore the only one that produces a stable response [1].

- True: If option 2 is used, the closed-loop system will have two poles to the right of -0.5, and a third left of -6. The last pole is much faster than the two close to the origin, so the response will be dominated by the complex pole pair. [2]

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln\left(\frac{k}{100}\sqrt{1-\zeta^2}\right)}{-\zeta\omega_n}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	<u>cosinusoid</u>
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped <u>cosinusoid</u>



Fig. 1 Krummel wearing a bowtie